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Black Body Radiation: Spectral Distribution

A black body is one which absorbs all the heat radiations incident on it. When radiations are allowed to fall on such a body they are neither reflected nor transmitted. A black body is called black body due to the fact that whatever may be the colour of the incident radiations, the body appears black. When such a body is heated, it emits radiations which we call as black body radiations. A cavity made out of hollow container of any material (iron or copper) with a narrow opening and painted with lampblack in the inside portion gives a close approximation to a perfectly black body. When any radiation falls on this hole, it enters the cavity, get reflected by the wall of the cavity and eventually gets absorbed. Now if we heat the container at various temperatures, it will emit radiations of all the frequencies (or wavelengths). So, the emitted radiation from a black body is continuous spectrum, i.e. it contains radiation of all frequencies.

Variation of Intensity of Black-Body Radiation

When a black body is heated at different temperatures, it emits radiations of all types of wavelengths. The origin of radiations from a heated body is the rapid vibrations of particles (known as oscillators) composing the body. According to Maxwell's electromagnetic theory, these oscillators emit radiant energy in the form of electromagnetic waves. Fig. 2.1 shows the variation

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of intensity of radiation with the wavelength at different temperatures.

From Fig. 2.1, the following conclusions can be drawn:

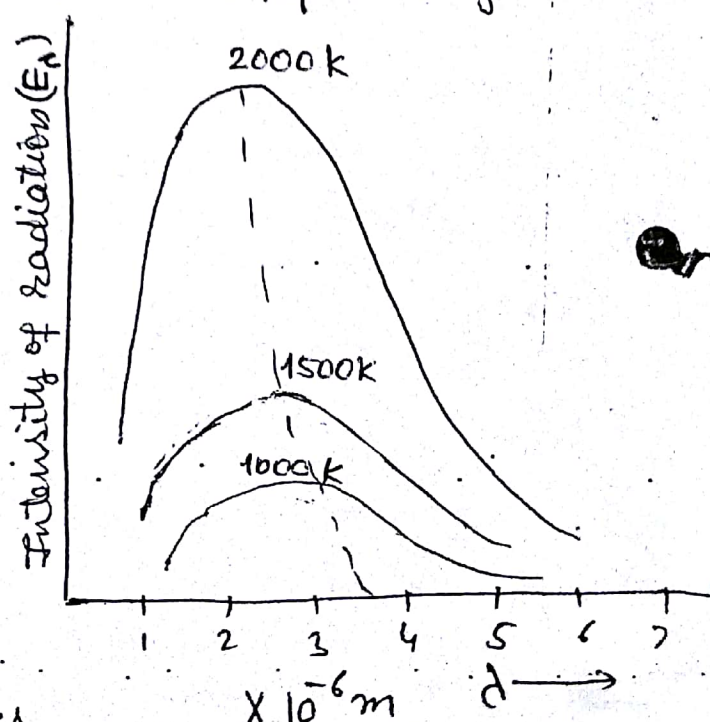
(i) At a given temperature, the energy is not uniformly distributed in the radiation spectrum of a hot body.

(ii) At a given temperature, the intensity of radiation increases with increase in wavelength and at a particular wavelength its value is maximum. With further increase in wavelength, the intensity of heat radiations decreases.

(iii) With increase in temperature, λ_m decreases such that $\lambda_m T = \text{Constant} = 0.2896 \text{ cm.K}$ (where λ_m is the wavelength at which maximum emission of energy takes place). This relation is called Wien's displacement law. The points on the dotted line are representing λ_m at various temperatures.

(iv) For all wavelengths, an increase in temperature causes an increase in the energy emission.

(v) The area under each curve represents the total energy emitted by the body at a particular temperature for the range of wavelengths considered. This area increases with increase of temperature.



It is found that the energy is directly proportional to the forth power of absolute temperature i.e., $E \propto T^4$ which is Stefan's law.

$$E = \int_0^\infty E_\lambda d\lambda = \sigma T^4$$

Difficulties in Explaining Black Body Spectrum:

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The classical electromagnetic theory or wave theory together with classical thermodynamics does not explain the characteristics of the blackbody spectrum. However, Planck's hypothesis can explain these characteristics together with the use of classical thermodynamics.

Classical wave theory says that the electromagnetic radiation inside the cavity of the blackbody at an equilibrium temperature T form the standing waves and the number of standing waves possible that can fit in the cavity depends on the wavelength. The number of possible modes in the cavity is large if the wavelength is small. However, for large wavelengths the number of possible modes is small. According to Rayleigh and Jeans, this increase in the number of modes is proportional to $1/\lambda^2$ or ν^2 and also each of the standing waves must be assigned an average kinetic energy kT , where k is the Boltzmann constant. This leads to the following Rayleigh-Jeans law

$$I(\nu) d\nu = \frac{8\pi\nu^2}{c^3} kT d\nu$$

This relation shows that $I(\nu)$ is proportional to the square of ν . The corresponding plot is shown in Fig 2.2. It is clear from the figure that the experimental data does not agree with the theory;

the agreement is good only for smaller values of ν . The disagreement at high frequencies, in the UV region, is called ultraviolet catastrophe. Thus, the distribution of a blackbody could not be explained on the basis of classical theory. This difficulty was resolved by Planck in 1900, when he stated that by assuming electromagnetic radiation to be emitted or absorbed in the form of

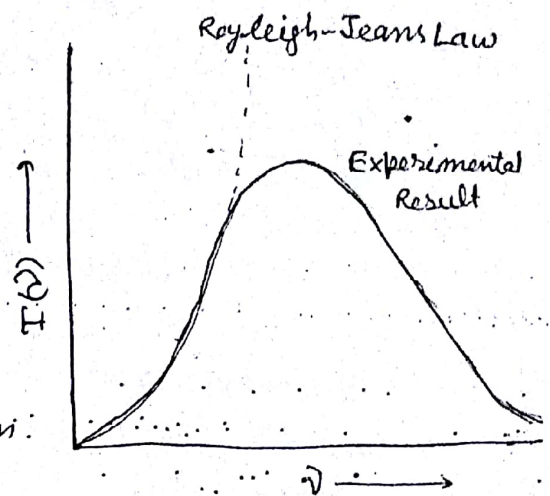


Fig. 2.2

the bundles of size $h\nu$, one could correctly predict the spectrum of black body radiation. As mentioned earlier, this bundle of energy is called a quantum. The quanta of high frequencies have high energies and those of low frequencies have low energies. Thus, the atom and molecules in the cavity will emit radiation only if they have energy in the excess of $h\nu$. For low frequencies ν , there will be a large number of atoms and molecules that might have this excess energy. Since the bundles become quite bigger for high frequencies ν . The number of atoms or molecules having energies in the excess of $h\nu$ decreases. It means for large ν , the intensity $I(\nu)$ does not increase rather decreases. For the explanation of black body radiation, Planck made a use of Maxwell-Boltzmann distribution. According to this distribution, the number of molecules N_n with energy E is given by

$$N_n = N_0 e^{-E/kT}$$

In the above expression, N_0 refers to the number of molecules in a system in equilibrium at temperature T .

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Planck

Planck combined the expression of N_m with quantum hypothesis $E = nh\nu$ and calculated the mean energy. Finally he arrived at the following expression for the distribution of the maximum intensity of radiation in the spectrum of black body.

$$I(\nu) = \frac{8\pi h \nu^3}{c^3} \cdot \frac{1}{e^{h\nu/KT} - 1}$$

This expression is referred to as Planck's radiation law. This theoretical formula fits very well with the experimental data for the entire wavelength as shown in Fig. 2.3. Thus Planck's quantum theory was able to interpret fully different characteristics of black body radiation with classical theory could not.

Planck's Quantum Hypothesis

Max Planck in 1900 introduced the quantum theory of radiation to explain the distribution of energy in the spectrum of blackbody radiation. He assumed that the atoms of the walls of a black body behave as oscillators and each has a characteristic frequency of oscillation. He made the following two revolutionary assumptions about the atomic oscillator.

(i) An oscillator cannot have any arbitrary value of energy, but can have only discrete energies as per the following relation.

$$E = n \cdot h \nu$$

where $n = 0, 1, 2, \dots$, ν and h are known as frequency of oscillation and Planck's constant ($= 6.62 \times 10^{-34}$ J. sec.), respectively. This relation shows that the energy of the oscillation is quantised.

(6)

(i2) The oscillator can emit or absorb energy only in the form of packets of energy $h\nu$, but not continuously. In other words, we can say that, the emission or absorption of energy occurs only when the oscillator jumps from one energy state to another along with the energy difference given by

$$\Delta E = \Delta n h\nu$$

or $E_2 - E_1 = (n_2 - n_1) h\nu$

Average Energy of Planck's Oscillators

If N be the total number of oscillators and E as the total energy of these oscillators, then the average energy will be given by the relation:

$$\bar{E} = \frac{E}{N} \quad \text{--- (1)}$$

Now consider $N_0, N_1, N_2, \dots, N_n$, as the number of oscillators having energy values $0, h\nu, 2h\nu, \dots, nh\nu$ respectively. Then by Maxwell's distribution formula $N_n = N_0 e^{-nh\nu/kT}$, we have

$$\begin{aligned} N &= N_0 + N_1 + N_2 + \dots \\ &= N_0 (1 + e^{-h\nu/kT} + e^{-2h\nu/kT} + \dots) \\ &= \frac{N_0}{(1 - e^{-h\nu/kT})} \quad \text{--- (2)} \end{aligned}$$

and the total energy is

$$\begin{aligned} E &= N_0 \times 0 + N_1 \times h\nu + N_2 \times 2h\nu + \dots \\ &= N_0 \times 0 + N_0 \times e^{-h\nu/kT} \cdot h\nu + N_0 \times e^{-2h\nu/kT} \cdot 2h\nu + \dots \\ &= N_0 e^{-h\nu/kT} h\nu [1 + 2e^{-h\nu/kT} + 3e^{-2h\nu/kT} + \dots] \\ &= N_0 e^{-h\nu/kT} h\nu \frac{1}{(1 - e^{-h\nu/kT})^2} \quad \text{--- (3)} \end{aligned}$$

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Putting the values of N and E from the eqs (2) and (3) eqn. (1), we get

$$\bar{E} = \frac{E}{N} = \frac{h\nu \bar{E} e^{-h\nu/kT}}{1 - e^{-h\nu/kT}} = \frac{h\nu}{e^{h\nu/kT} - 1} \quad (4)$$

This is the ^{expression for} average energy of a Planck's oscillator.

Planck's Radiation Formula

The energy density of radiation (u_ν) in the frequency range ν and $\nu + d\nu$ depending upon the average energy of an oscillator is given by

$$u_\nu d\nu = \frac{8\pi\nu^2}{c^3} d\nu \times \bar{E} \quad (5)$$

Putting the value of $\bar{E}$ from eq. (4) in eq. (5), gives

$$u_\nu d\nu = \frac{8\pi\nu^2}{c^3} \cdot \frac{h\nu}{e^{h\nu/kT} - 1} d\nu$$

$$\text{or } u_\nu d\nu = \frac{8\pi h}{c^3} \frac{\nu^3 d\nu}{e^{h\nu/kT} - 1} \quad (6)$$

The above relation is known as Planck's radiation formula in terms of frequency. This law can also be expressed in terms of wavelength λ of the radiation. This law can also be expressed in terms of wavelength λ of the radiation.

Since $\nu = \frac{c}{\lambda}$ for electromagnetic radiation, $d\nu = -\frac{c}{\lambda^2} d\lambda$. Further, we know that the frequency is reciprocal of wavelength or in other words an increase in frequency corresponds to a decrease in wavelength. Therefore

$$u_\lambda d\lambda = -u_\nu d\nu$$

$$\text{or } u_\lambda d\lambda = -\frac{8\pi h}{c^3} \cdot \frac{\left(\frac{c}{\lambda}\right)^3 \left(-\frac{c}{\lambda^2} d\lambda\right)}{e^{hc/\lambda kT} - 1} = \frac{8\pi h c}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1} d\lambda \quad (7)$$

The above relation is known as Planck's formula in terms of wavelength.

Wein's Law and Rayleigh-Jeans Law

(8)

With the help of Planck's radiation formula, Wein's law and Rayleigh-Jeans law can be derived. When the wavelength λ and temperature T are very small, then $e^{hc/\lambda kT} \gg 1$. Therefore, 1 can be neglected in the denominator of eq (7). Thus

$$u_{\lambda} d\lambda = \frac{8\pi hc}{\lambda^5} e^{-hc/\lambda kT} d\lambda$$

By substituting $8\pi hc = A$ and $\frac{hc}{k} = B$, we get

$$u_{\lambda} d\lambda = \frac{A}{\lambda^5} e^{-B/\lambda T} d\lambda \quad \text{--- (8)}$$

This is known as Wein's law, which is valid at low temperature T and small wavelength λ .

For high temperature T and large wavelength λ , $e^{hc/\lambda kT}$ can be approximated to $1 + \frac{hc}{\lambda kT}$. Then we have from Eq (7)

$$u_{\lambda} d\lambda = \frac{8\pi hc}{\lambda^5 \left[1 + \frac{hc}{\lambda kT} - 1 \right]} d\lambda$$

$$\boxed{u_{\lambda} d\lambda = \frac{8\pi kT}{\lambda^4} d\lambda}$$

This is known as Rayleigh-Jeans law.

Statistical Mechanics

To study of statistical mechanics mainly consists of two branches, namely classical statistics and quantum statistics. In order to discuss both these statistics we shall first discuss certain physical terms; e.g. static system, dynamic system, position space, momentum space, phase space and volume in phase space.

A system, in which its constituent particles are at rest, is called static system, if the constituent particles of the system are in constant state of motion, the system is called dynamic system.

The position of a particle in a three dimensional static system is completely defined by three position coordinates x, y, z . This three dimensional space is called the position space.

In dynamic system, the particles move with various velocities and possess momenta. Let p_x, p_y and p_z be the components of the momentum along three coordinates. Thus the momentum, p , of the particle is completely specified by three momentum coordinates p_x, p_y, p_z defines volume Γ associated with momentum space. The product of these two volumes (V and Γ), is called available volume in phase space and represented by τ . Thus, the elementary volume (small cell) $d\tau$ in phase space is given by

$$d\tau = dx dy dz dp_x dp_y dp_z = h^3$$

where h is a quantity having dimensions of action. Classically, the position and momentum of the

particle can be measured by any degree of accuracy and the volume $d\tau$ or h^3 can be made small, approaching zero. But in quantum mechanics, h cannot be made as small as we want.

The value of h is determined by Heisenberg uncertainty principle, i.e.

$$dx \cdot dp_x \geq h$$

$$dy \cdot dp_y \geq h$$

and

$$dz \cdot dp_z \geq h$$

where h is Planck's Constant of action. Thus

$$d\tau = dx \cdot dy \cdot dz \cdot dp_x \cdot dp_y \cdot dp_z$$

$$d\tau \geq h^3$$

This shows that according to uncertainty principle, the volume of elementary cell in phase space is greater than or equal to h^3 . In other words h^3 is the minimum volume of a cell in phase space.

Types of Statistics:

The available volume in phase space is divided into large number of compartments and each compartment is further divided into large number of elementary cells of equal size. The distribution of particles depends upon:

- (i) whether the particles are distinguishable or indistinguishable.
- (ii) the total number of cells or compartments is determined by the minimum allowed volume of each cell and

(ii) whether there is any restriction on the number of particles that can go in any cell.

Based on criteria (i) and (ii) there are two types of statistics: (a) Classical statistics and (b) quantum statistics. These statistics have been distinguished on the following features:

(i) The particles in classical statistics are considered distinguishable, while in quantum statistics the particles are treated as indistinguishable.

(ii) In classical statistics, the volume of elementary cell in phase space can be made as small as we want, while in quantum statistics, its value cannot be less than h^3 , where h is Planck's Constant.

Classical Statistics

In classical statistics, there is only one type of statistics, called Maxwell-Boltzmann statistics. In this statistics particles are treated as distinguishable and any number of them can occupy a single cell in phase space.

According to Maxwell-Boltzmann statistics

$$n_i = \frac{g_i}{e^{\alpha + \beta \epsilon_i}}$$

This is called Maxwell-Boltzmann distribution law. It is applicable to an ideal gas molecules.

① Quantum Statistics: There are two types of quantum statistics

- (i) Bose-Einstein statistics
- (ii) Fermi-Dirac statistics

In Bose-Einstein (B-E) statistics, particles are treated as indistinguishable and any number of the particles can occupy a single cell in phase space. It applies to particles having integral spin angular momentum e.g. photons (spin 1), mesons, α particles (spin zero) etc. and such particles are called Bosons.

In Fermi-Dirac (F-D) statistics, the particles are indistinguishable, but obey Pauli exclusion principle, according to which no two particles can occupy the same cell in phase space i.e. not more than one particle can occupy a single cell. Particles obeying F-D statistics have half-integral spin angular momentum and are called Fermions. Such particles are protons, electrons, neutrons etc. all these particles having spin $1/2$.

Bose-Einstein Statistics

According to this statistics the most probable distribution of the particles among various energy levels for a system obeying Bose-Einstein statistics

$$n_i = \frac{g_i}{e^{\alpha + \beta \epsilon_i} - 1}$$

Fermi-Dirac Statistics

According to this statistics the most probable of particles among various energy levels obeying Fermi-Dirac Statistics

$$n_i = \frac{g_i}{e^{\alpha + \beta \epsilon_i} + 1}$$

Where α and β are constants and can be obtained for all three distributions. It can be shown that $\beta = \frac{1}{kT}$, where k is Boltzmann constant and T absolute temperature. The value of $\alpha = -\frac{E_F}{kT}$.

Here we define $F(\epsilon_i) = n_i/g_i$ called occupation index of state of energy ϵ_i . It represents the average number of particles in each of the states of that energy. In case of photon gas for which $\alpha = 0$ and $E = h\nu$.

Difference Between Three Distribution Laws

The three distribution functions i.e. Maxwell-Boltzmann, Bose-Einstein and Fermi-Dirac are expressed as

(i) Maxwell-Boltzmann function

$$\frac{n_i}{g_i} = \frac{1}{e^{\alpha} e^{\epsilon_i/kT}} = F_{MB} \quad \text{--- (1)}$$

(ii) Bose-Einstein function

$$\frac{n_i}{g_i} = \frac{1}{e^{\alpha} e^{\epsilon_i/kT} - 1} = F_{BE} \quad \text{--- (2)}$$

(iii) Fermi-Dirac function

$$\frac{n_i}{g_i} = \frac{1}{e^{\alpha} e^{\epsilon_i/kT} + 1} = F_{FD} \quad \text{--- (3)}$$

(14) Fig. 2.4 shows the plot of Maxwell-Boltzmann distribution function F_{MB} . It falls exponentially with increase in energy E_i . It falls by a factor $(1/e)$ for each increase of kT in the energy.

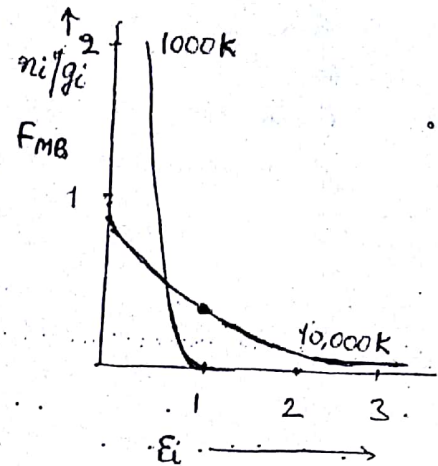


Fig. 2.4

Fig. 2.5 shows the Bose-Einstein distribution function for two temperatures i.e., $T_1 = 1000 \text{ K}$ and $T_2 = 10,000 \text{ K}$ for $\alpha = 0$.

It can be observed that if $e^{\alpha} e^{E_i/kT} \gg 1$ or $g_i/n_i \gg 1$, then Bose-Einstein and Fermi-Dirac statistics show same result as obtained by Maxwell-Boltzmann statistics.

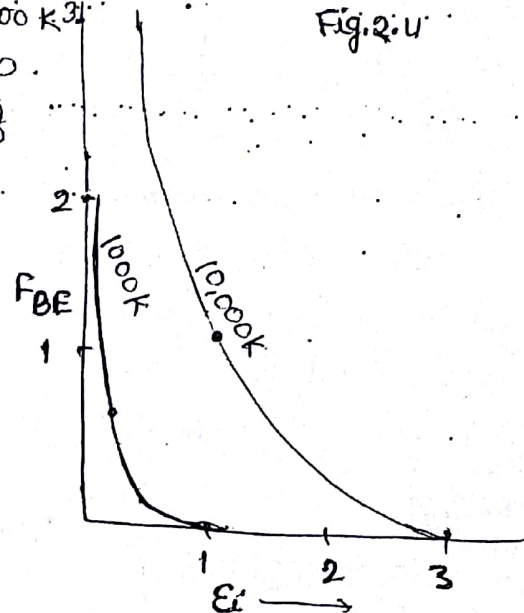


Fig. 2.5

Fermi-Dirac Statistics:

This statistics is applicable to system, which consist of identical, independent and indistinguishable particles of having half-integral spins. The particles, which obey Fermi-Dirac statistics, are called fermions. The examples of fermions are electrons, protons, neutrons etc. The fermions must obey Pauli exclusion principle. In Fermi-Dirac statistics, interchange of two particles of the system leaves the resultant system in an antisymmetric state. That is, the wave function of the system gets changed only with minus sign. As it obeys the Pauli exclusion principle, in Fermi-Dirac statistics, there

can be only one particle in each state. Hence, the total number of particles must be less than or equal to the total number of states available. Under these considerations, fermions lead to the following distribution law, named Fermi-Dirac distribution law, given by

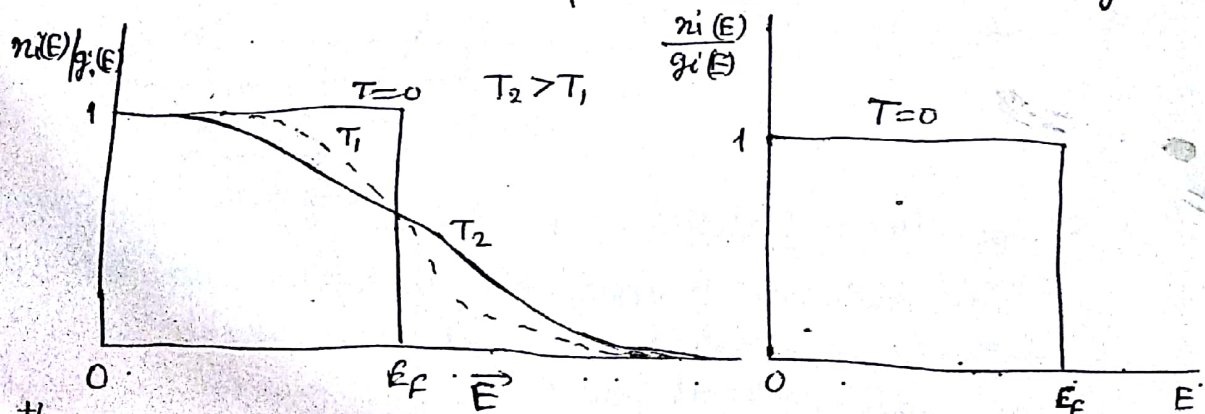
$$n_i(E) = \frac{g_i(E)}{e^{\alpha + \beta E} + 1} \quad \text{--- (1)}$$

where $\alpha = -E_F/kT$, $\beta = 1/kT$

$$\text{So } n_i(E) = \frac{g_i(E)}{e^{(E-E_F)/kT} + 1} \quad \text{--- (2)}$$

$$\frac{n_i(E)}{g_i(E)} = \frac{1}{e^{(E-E_F)/kT} + 1} \quad \text{--- (3)}$$

In the above equations, n_i is the number of particles in an energy state E , g_i is the statistical weight factor and E_F is the fermi energy. Fermi energy is independent of temperature. The plots of $n_i(E)/g_i(E)$ versus E for different temperatures is shown in Fig. 2.6



Now, the different cases will be discussed.

(a) At $T=0$ and when $E < E_F$. Eq. (3)

$$\frac{n_i(E)}{g_i(E)} = 1 \quad \text{for } E < E_F \quad \left\{ \because e^{\infty} = 0 \right\}$$

It means that $n_i(E) = g_i(E)$, i.e. all the energy state will have one electron each.

$$= 0 \quad \text{for } E > E_F \quad \left\{ \because e^{\infty} = \infty \right\}$$

It means that $n_i = 0$, i.e. all the energy state which have energies greater than fermi energy are vacant. This

- (16) clarifies that all states with energies up to E_F are filled while all states with energy greater than E_F are vacant. The plot between $n_i(E)/g_i(E)$ with E is shown in Fig. 15.11 for these conditions at $T = 0K$. Comparison of M.B., B-E and F-D Statistics

Maxwell-Boltzmann Statistics (Classical Statistics)	Bose-Einstein Statistics (Quantum Statistics)	Fermi-Dirac Statistics (Quantum Statistics)
1. Apply to system of distinguishable particles such as gases.	Apply to systems of indistinguishable particles not obeying Pauli's Exclusion principle, such as photons, phonons and liquid helium at low temp.	Apply to system of indistinguishable particles obeying Pauli exclusion principle, such as free electrons in metal, electrons in a star.
2. The distribution law is $n_i = \frac{g_i}{e^{\alpha} e^{E_i/kT}}$	The distribution law is $n_i = \frac{g_i}{e^{\alpha} e^{E_i/kT} - 1}$	The distribution law is $n_i = \frac{g_i}{e^{\alpha} e^{E_i/kT} + 1}$
3. There is no restriction on the number of particles in the given state.	There is no restriction on the number of particles in a given state.	only one particle in a given quantum state is allowed.
4. In the phase space the volume of a phase cell is not fixed.	The volume of phase cell is of the order of h^3 .	The volume of phase cell is of the order of h^3 .
5. The behaviour of distribution function $f(E_i)$ against E_i is exponential.	For $E_i \gg kT$, exponential - for $E_i \ll kT$, lies above M.B.	For $E_i \gg kT$, exponential where $E_i \gg E_F$. If $E_F \gg kT$ decreases abruptly near E_F .
6. The energy of an M.B. system, such as in ideal monochromatic gas, may be zero at absolute zero.	The energy may be zero at absolute zero.	The energy at absolute zero can not be zero because all particles cannot come down to ground state due to Pauli's exclusion principle.